PUNJAB ENGINEERING COLLEGE (DEEMED TO BE UNIVERSITY)
Sector 12, Chandigarh - 160012 | Office of Dean Academic Affairs (DAA)
Capstone Coursework Deliverable • Academic Clearance Automation Engine

Unified Modeling Language (UML) Structural & Behavioral Models

Academic Grievance, No-Dues & Clearance Automation Engine

Institution: Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh - 160012
Document Identifier: PEC-SE-UML-2026-FROZEN
Standard Compliance: OMG UML 2.5 Specification
Baseline Version: 3.0.0 (FROZEN ARCHITECTURAL BASELINE)
Administrative Authority: Office of the Dean Academic Affairs (DAA), PEC Chandigarh
Status: FROZEN & RATIFIED (Implementation-Agnostic Baseline)

Baseline Freeze Declaration

> **NOTICE OF BASELINE FREEZE:**
> This specification models the complete, immutable object-oriented architecture, behavioral sequences, state lifecycles, and component interfaces for the PEC Clearance & Grievance Automation Engine.
>
> The use case actors, class schemas, interface contracts, state transitions, and interaction sequences modeled in this document are **FROZEN**. Internal implementation refactoring (e.g., helper function reorganizations or styling adjustments) shall conform strictly to these structural and behavioral models without requiring modifications to this document.

OMG UML 2.5 Modeling Rules & Standards Compliance Matrix

This document conforms strictly to the **Object Management Group (OMG) UML 2.5 Superstructure Specification**:

Diagram Type	Diagram Classification	Formal OMG UML 2.5 Rules & Semantics Applied	Section
Use Case Diagram	Behavioral	Actors outside system boundary; use cases within boundary; precise `<<include>>` (mandatory functional dependency) and `<<extend>>` (conditional extension) stereotypes.	Section 1
Class Diagram	Structural	3-compartment notation; member visibility indicators (`+` Public, `#` Protected, `-` Private); precise relationship semantics: Generalization (`<	`--`), `Realization` (..
Activity Diagram	Behavioral	Concurrency fork and join bars; decision and merge diamonds; action states; initial/final activity nodes; swimlane encapsulation of gate authorities.	Section 3
Sequence Diagram	Interaction	Participant lifelines; strictly descending chronological time axis; synchronous calls (`->>`), returns (`-->>`); interaction use fragments (`alt` , `opt` , `loop`).	Section 4
State Machine Diagram	Behavioral	State transitions triggered by events and guards (`Event [guard] / action`); composite states (`Gate 1` , `Gate 2` , `Gate 3`); initial pseudo-states (`[*]`).	Section 5
Component & Deployment	Structural / Physical	3-tier boundary separation (Client UI → Application Services → Persistent Data Store); interface connectors and protocol definitions.	Section 6

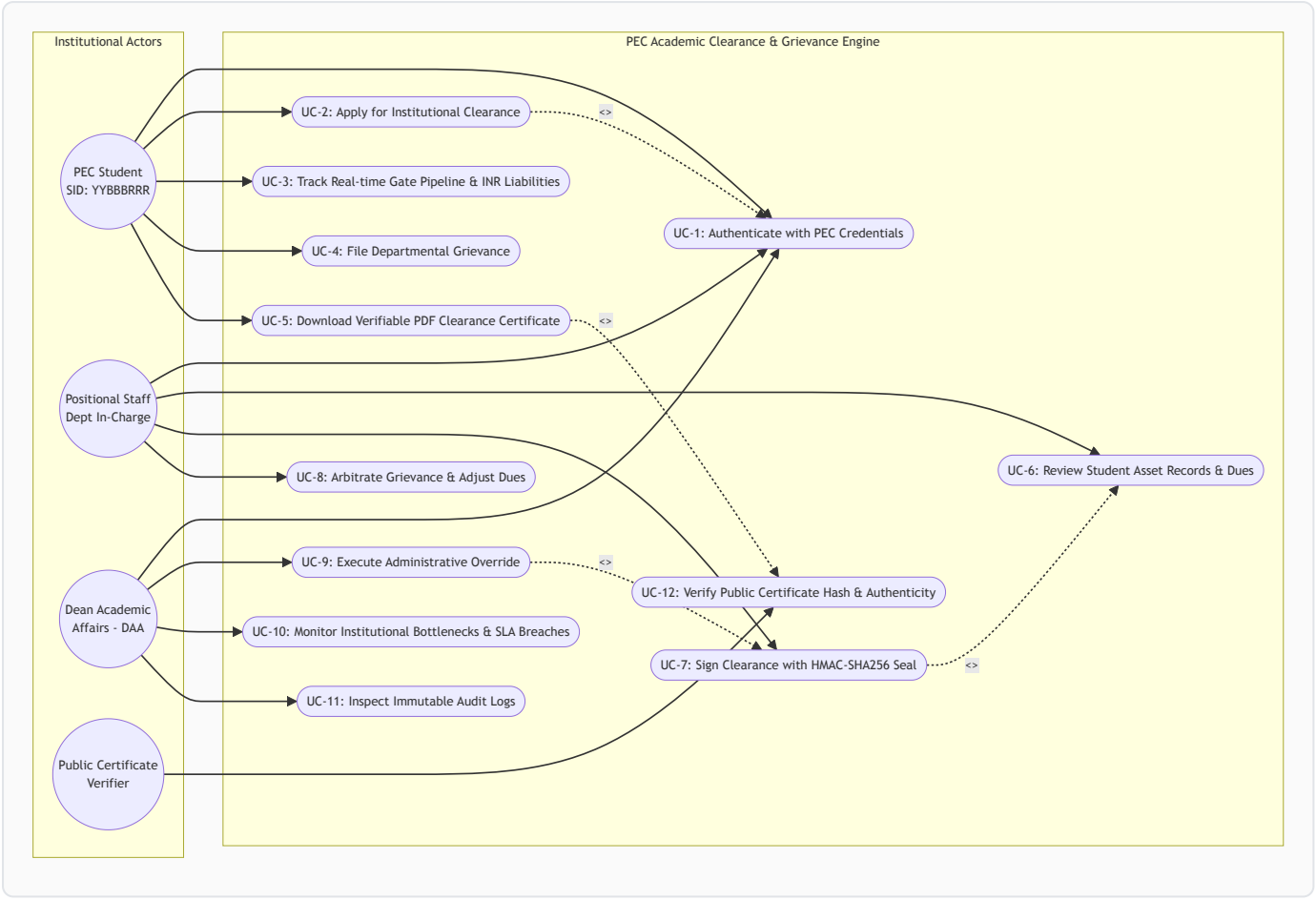
---

1. Use Case Model & Diagram

1.1 Actors & Role Boundaries

- **Student Actor:** Registered PEC student identified by an 8-digit SID (e.g., `22103042`).
- **Positional Staff Actor:** Departmental authority operating under permanent institutional designations (*Librarian*, *Chief Warden*, *Gymkhana President*, *HOD Labs*, *Accounts Officer*).
- **Dean Academic Affairs (DAA / Admin Actor):** Executive authority holding override privileges and institutional audit oversight.
- **Public Certificate Verifier Actor:** External unauthenticated party validating clearance credentials.

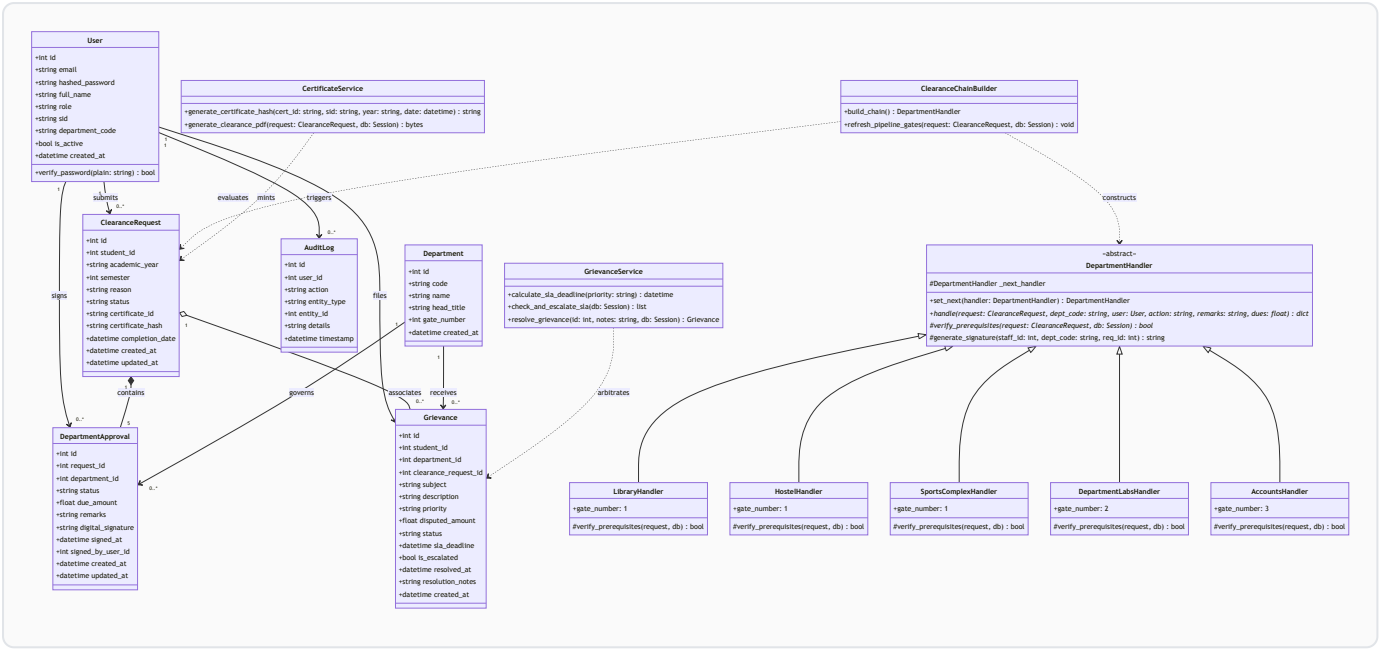
1.2 Use Case Diagram



---

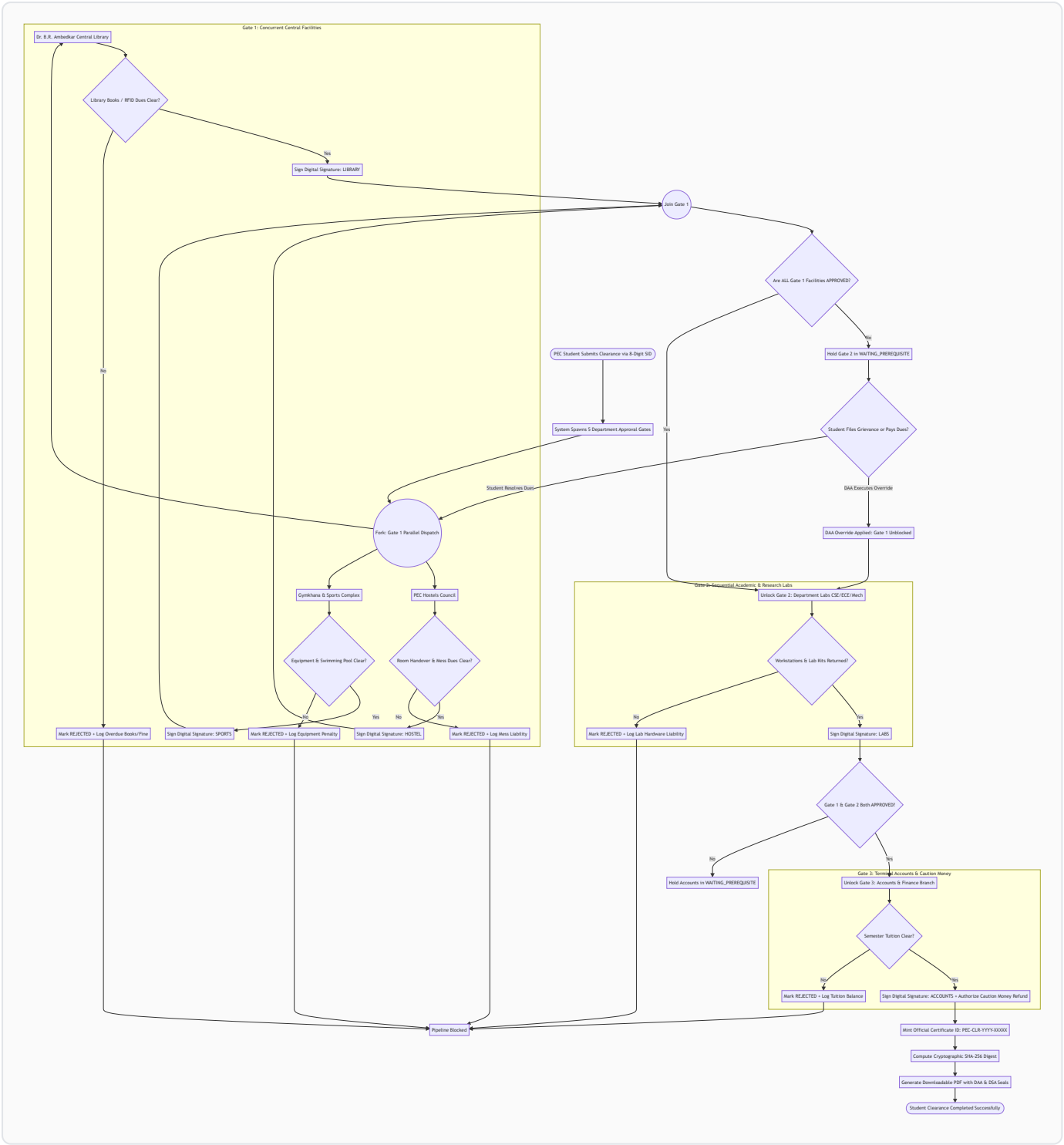
2. Static Structural Model: Class Diagram

The class diagram captures the object-oriented domain entities, the Chain of Responsibility behavioral pattern handlers, cryptographic helper utilities, and service orchestration layers:



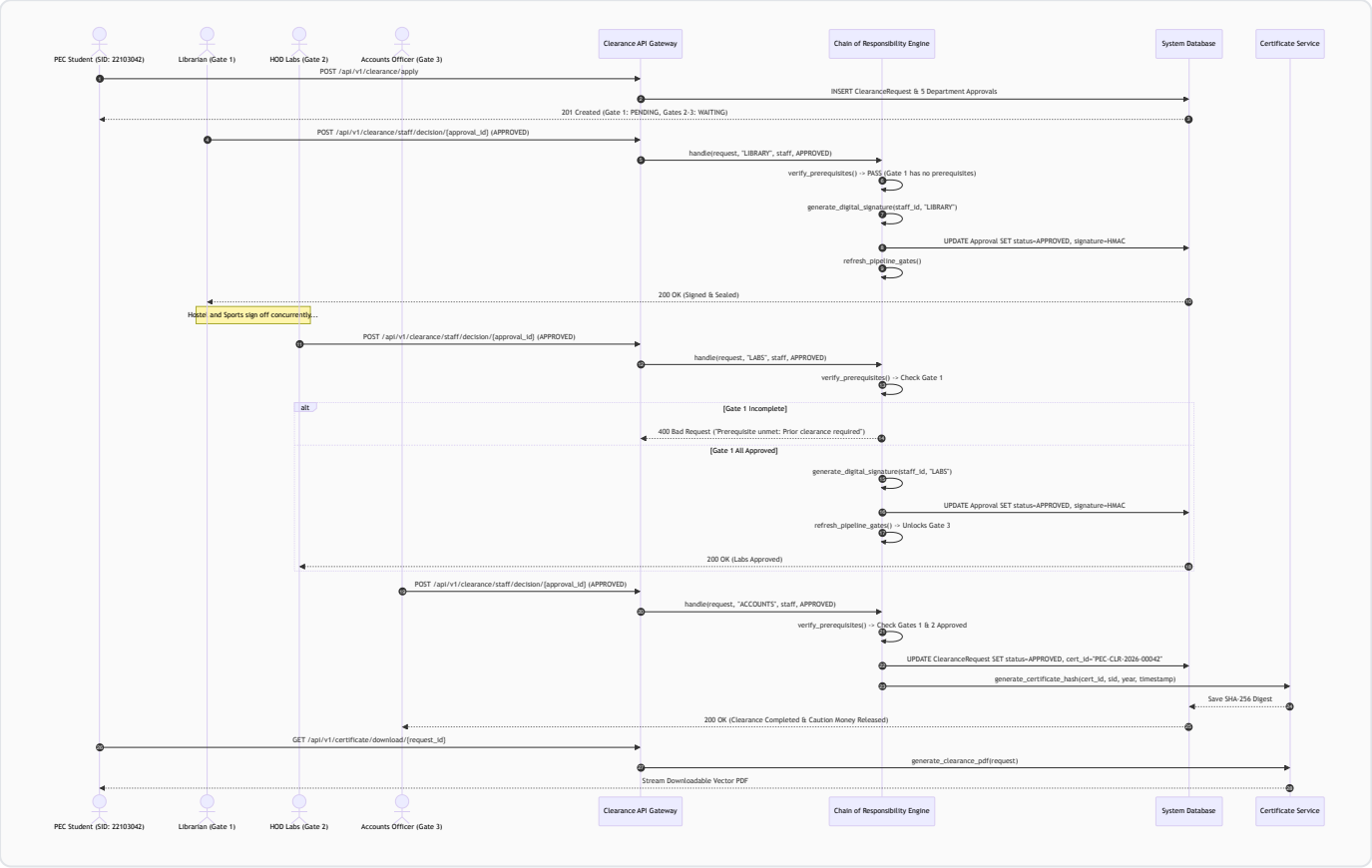
3. Dynamic Behavioral Model: Activity Diagram

This activity diagram captures the end-to-end lifecycle of a PEC clearance request, illustrating parallel concurrency forks, barrier joins, and sequential dependency gates:

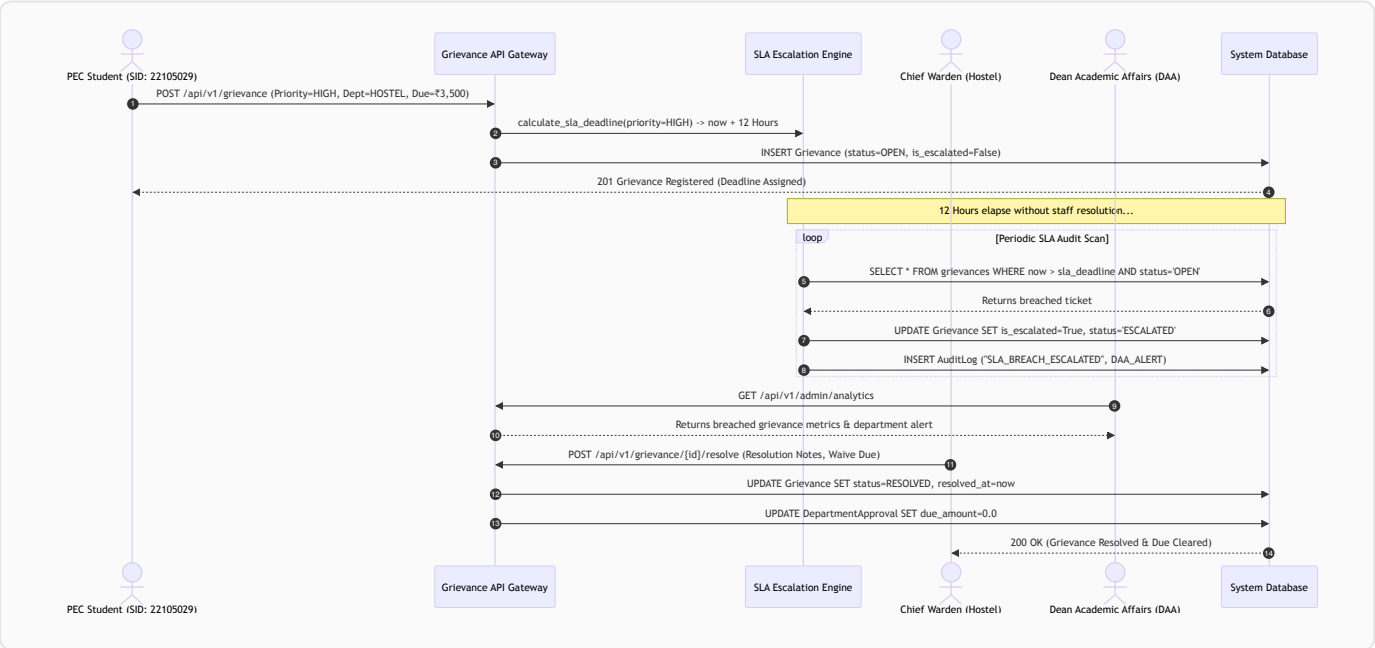

Gate 1: Concurrent Central Facilities
Dr. B.R. Ambedkar Central Library
Library Books / RFID Dues Clear?
Yes
No
Sign Digital Signature: LIBRARY
Mark REJECTED + Log Overdue Books/Fine
Join Gate 1
Are ALL Gate 1 Facilities APPROVED?
Yes
No
PEC Student Submits Clearance via 8-Digit SID
System Spawns 5 Department Approval Gates
Hold Gate 2 in WAITING_PREREQUISITE
Student Files Grievance or Pays Dues?
Student Resolves Dues
DAA Executes Override
DAA Override Applied: Gate 1 Unblocked
Fork: Gate 1 Parallel Dispatch
Gymkhana & Sports Complex
PEC Hostels Council
Equipment & Swimming Pool Clear?
Room Handover & Mess Dues Clear?
Yes
No
Yes
No
Sign Digital Signature: SPORTS
Mark REJECTED + Log Equipment Penalty
Sign Digital Signature: HOSTEL
Mark REJECTED + Log Mess Liability
Gate 2: Sequential Academic & Research Labs
Unlock Gate 2: Department Labs CSE/ECE/Mech
Workstations & Lab Kits Returned?
No
Yes
Mark REJECTED + Log Lab Hardware Liability
Sign Digital Signature: LABS
Gate 1 & Gate 2 Both APPROVED?
No
Yes
Hold Accounts in WAITING_PREREQUISITE
Gate 3: Terminal Accounts & Caution Money
Unlock Gate 3: Accounts & Finance Branch
Semester Tuition Clear?
No
Yes
Mark REJECTED + Log Tuition Balance
Sign Digital Signature: ACCOUNTS + Authorize Caution Money Refund
Pipeline Blocked
Mint Official Certificate ID: PEC-CLR-YYYY-XXXXX
Compute Cryptographic SHA-256 Digest
Generate Downloadable PDF with DAA & DSA Seals
Student Clearance Completed Successfully

4. Interaction Models: Sequence Diagrams

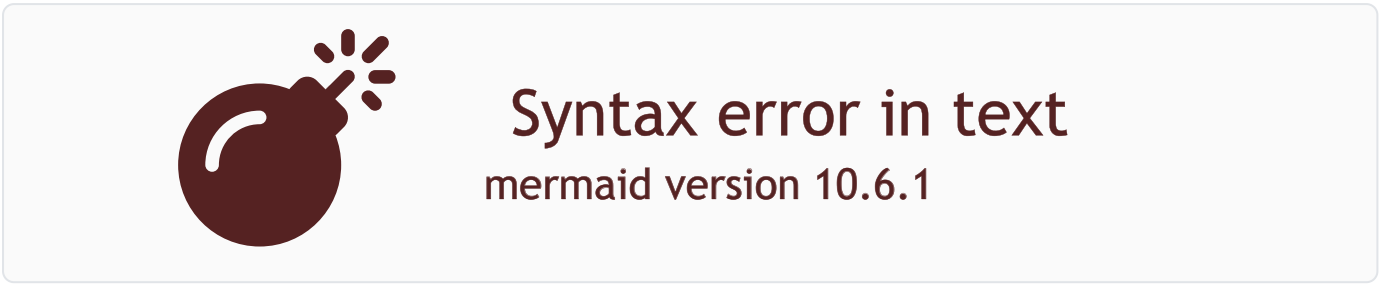
4.1 Sequence Diagram 1: Multi-Gate Approval & Cryptographic Minting



4.2 Sequence Diagram 2: Grievance Lifecycle & Automated DAA Escalation



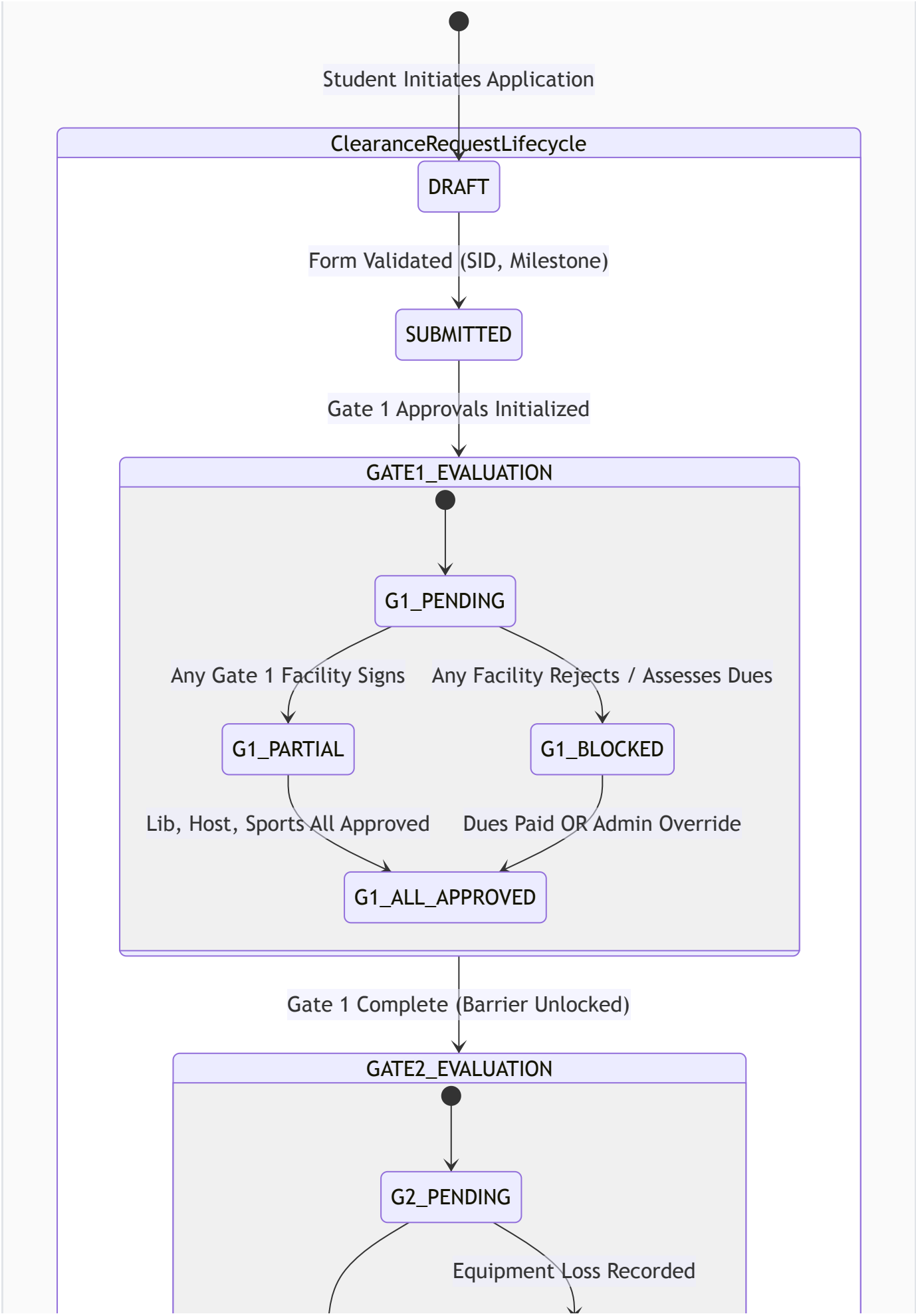
4.3 Sequence Diagram 3: Administrative Override & Concurrency Race Prevention

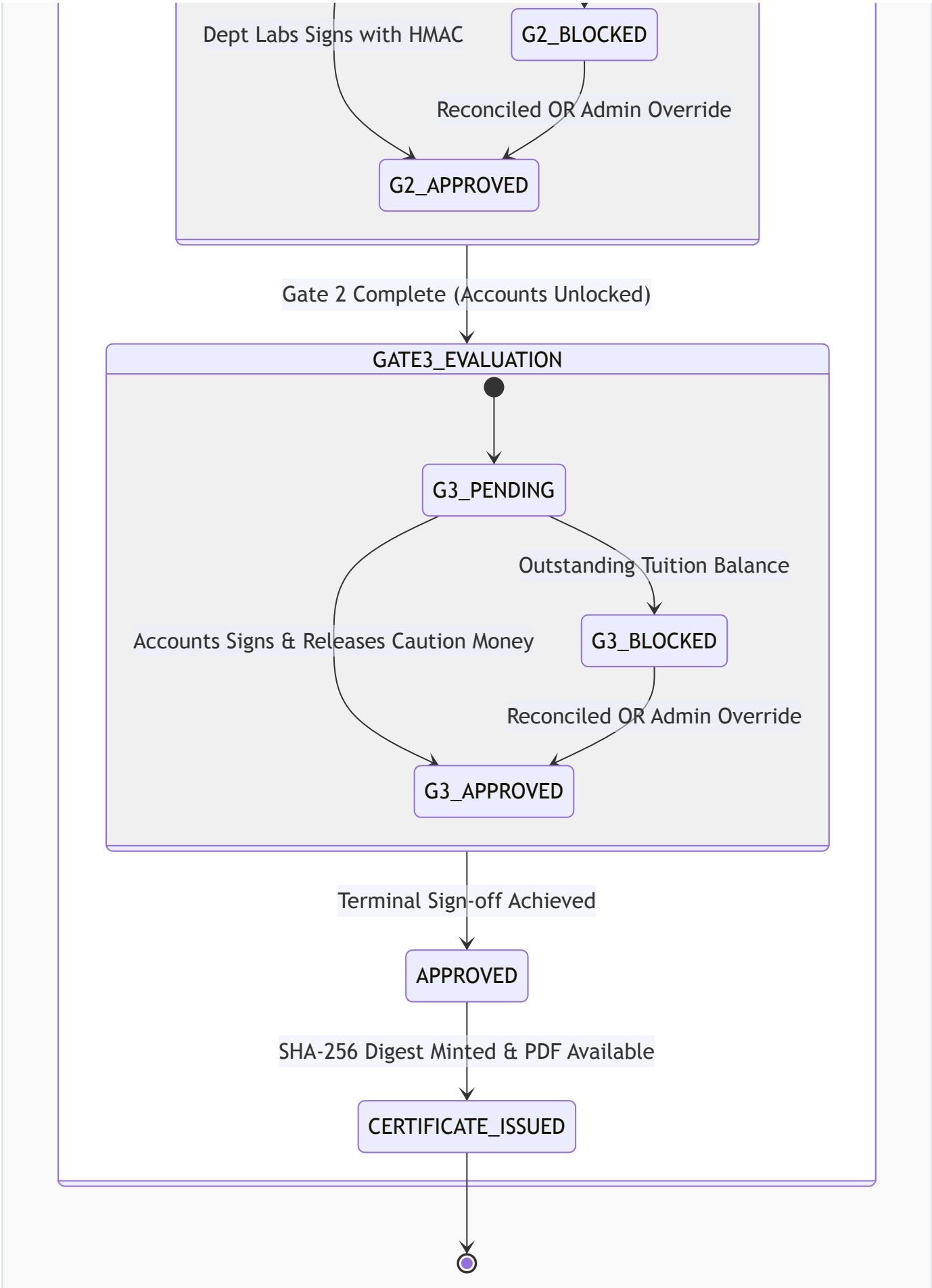


---

5. Behavioral State Machine Diagram

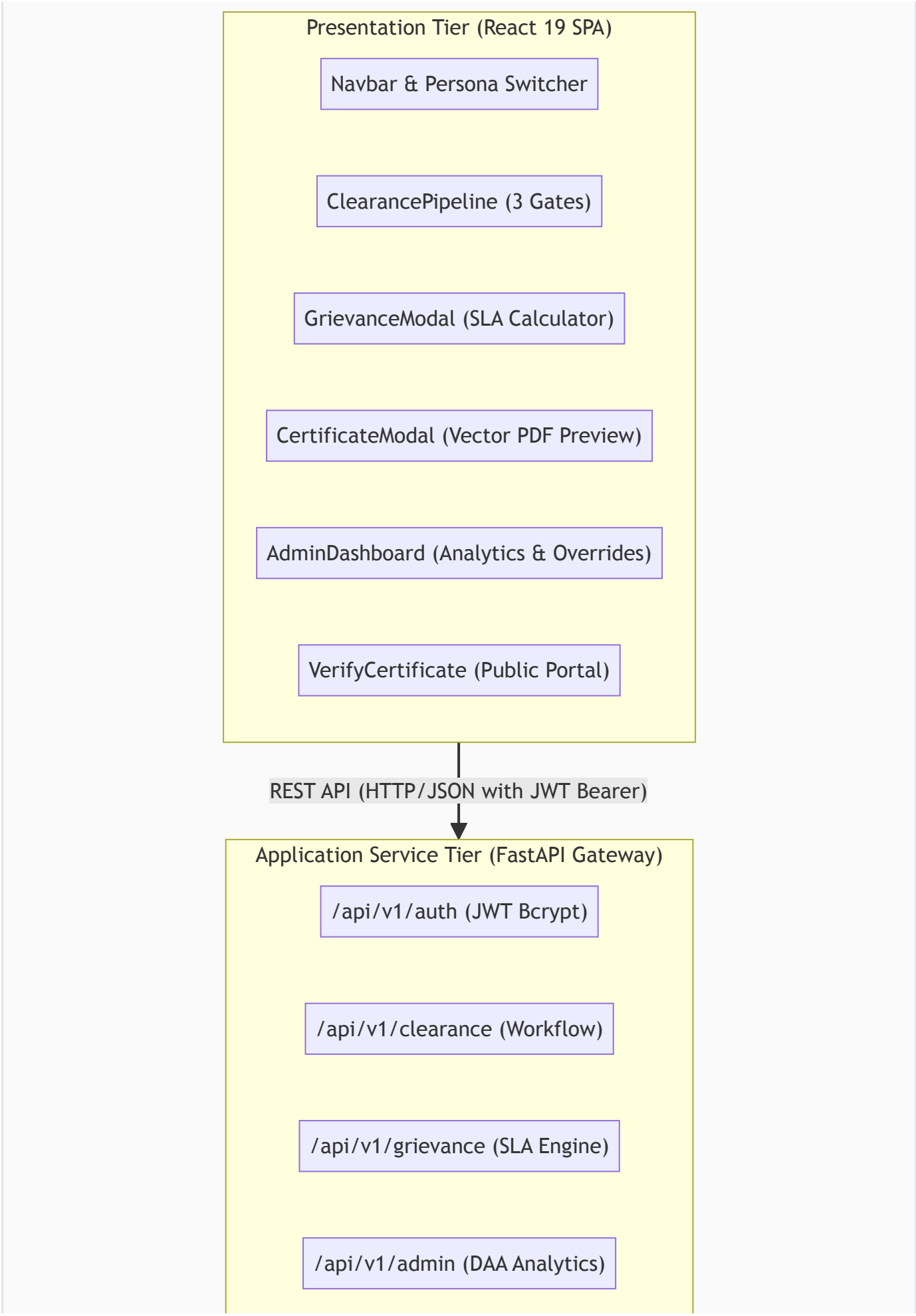
This state machine models the discrete status lifecycles of both the master clearance request and individual department approval gates:

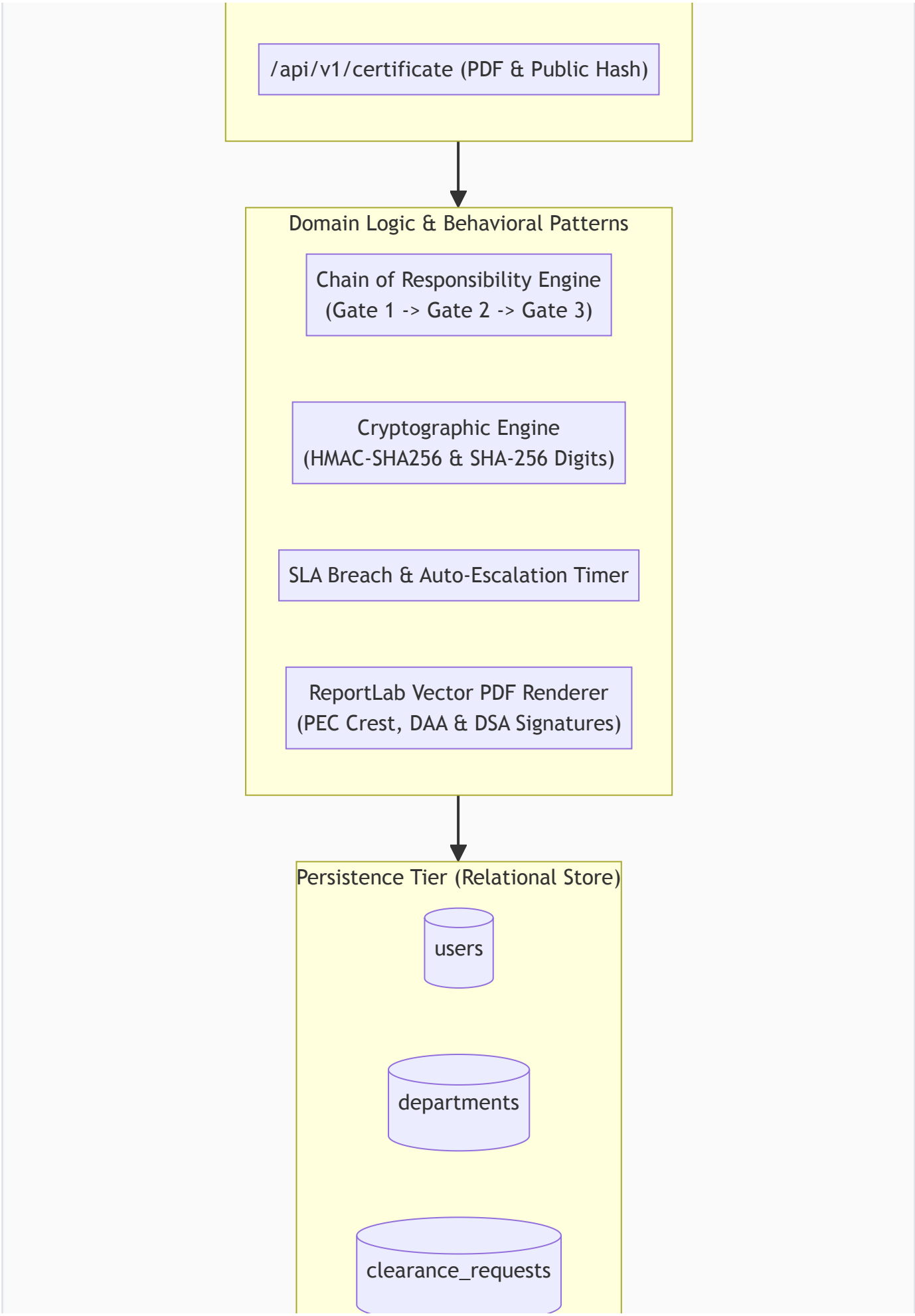

Student Initiates Application
ClearanceRequestLifecycle
DRAFT
Form Validated (SID, Milestone)
SUBMITTED
Gate 1 Approvals Initialized
GATE1_EVALUATION
G1_PENDING
Any Gate 1 Facility Signs
Any Facility Rejects / Assesses Dues
G1_PARTIAL
G1_BLOCKED
Lib, Host, Sports All Approved
Dues Paid OR Admin Override
G1_ALL_APPROVED
Gate 1 Complete (Barrier Unlocked)
GATE2_EVALUATION
G2_PENDING
Equipment Loss Recorded

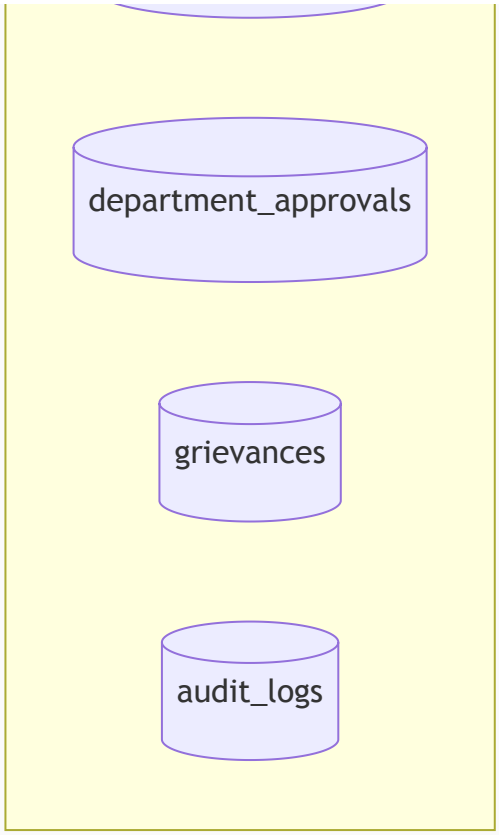

Dept Labs Signs with HMAC
G2_BLOCKED
Reconciled OR Admin Override
G2_APPROVED
Gate 2 Complete (Accounts Unlocked)
GATE3_EVALUATION
G3_PENDING
Outstanding Tuition Balance
Accounts Signs & Releases Caution Money
G3_BLOCKED
Reconciled OR Admin Override
G3_APPROVED
Terminal Sign-off Achieved
APPROVED
SHA-256 Digest Minted & PDF Available
CERTIFICATE_ISSUED

6. System Architecture & Component Deployment Diagram

The component diagram defines the physical and logical boundaries of the system, verifying complete separation between presentation, application services, and persistent storage:


Presentation Tier (React 19 SPA)
Navbar & Persona Switcher
ClearancePipeline (3 Gates)
GrievanceModal (SLA Calculator)
CertificateModal (Vector PDF Preview)
AdminDashboard (Analytics & Overrides)
VerifyCertificate (Public Portal)
REST API (HTTP/JSON with JWT Bearer)
Application Service Tier (FastAPI Gateway)
/api/v1/auth (JWT Bcrypt)
/api/v1/clearance (Workflow)
/api/v1/grievance (SLA Engine)
/api/v1/admin (DAA Analytics)


/api/v1/certificate (PDF & Public Hash)
Domain Logic & Behavioral Patterns
Chain of Responsibility Engine
(Gate 1 -> Gate 2 -> Gate 3)
Cryptographic Engine
(HMAC-SHA256 & SHA-256 Digits)
SLA Breach & Auto-Escalation Timer
ReportLab Vector PDF Renderer
(PEC Crest, DAA & DSA Signatures)
Persistence Tier (Relational Store)
users
departments
clearance_requests


department_approvals
grievances
audit_logs